

# Lecture 3: Simple Linear Regression and OLS

Economics 326 — Introduction to Econometrics II

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## Introduction and Definitions

### Overview of Simple Linear Regression

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- The **simple linear regression model** is used to study the relationship between **two variables**.
- It has many limitations, but nevertheless there are examples in the literature where the simple linear regression is applied (e.g., stock returns predictability).
- It is also a good starting point to learning the regression technique.

### Key Definitions

#### Terminology for Simple Regression

| $y$                | $x$                  |
|--------------------|----------------------|
| Dependent variable | Independent variable |
| Explained variable | Explanatory variable |
| Response variable  | Control variable     |
| Predicted variable | Predictor variable   |
| Regressand         | Regressor            |

### Sample vs Population

- The econometrician observes random data:

| observation | dependent variable | regressor |
|-------------|--------------------|-----------|
| 1           | $Y_1$              | $X_1$     |
| 2           | $Y_2$              | $X_2$     |
| $\vdots$    | $\vdots$           | $\vdots$  |
| $n$         | $Y_n$              | $X_n$     |

- A pair  $X_i, Y_i$  is called an observation.
- **Sample:**  $\{(X_i, Y_i) : i = 1, \dots, n\}$ .
- The population is the **joint distribution** of the sample.

# Simple Linear Regression Model

## Conditional Expectation Function

- We model the relationship between  $Y$  and  $X$  using the **conditional expectation**:

$$E[Y_i|X_i] = \alpha + \beta X_i.$$

- **Intercept:**  $\alpha = E[Y_i|X_i = 0]$ .
- **Slope:**  $\beta$  measures the effect of a unit change in  $X$  on  $Y$ :

$$\begin{aligned}\beta &= E[Y_i|X_i = x + 1] - E[Y_i|X_i = x] \\ &= [\alpha + \beta(x + 1)] - [\alpha + \beta x].\end{aligned}$$

- **Marginal effect** of  $X$  on  $Y$ :

$$\beta = \frac{dE[Y_i|X_i]}{dX_i}.$$

- The effect is the same for all  $x$ .

## Regression Error Term

- $\alpha$  and  $\beta$  in  $E[Y_i|X_i] = \alpha + \beta X_i$  are **unknown**.
- **Residual (error):**

$$U_i = Y_i - E[Y_i|X_i] = Y_i - (\alpha + \beta X_i).$$

$U_i$ 's are **unobservable**.

- **The model:**

$$\begin{aligned}Y_i &= \alpha + \beta X_i + U_i, \\ E[U_i|X_i] &= 0.\end{aligned}$$

## Functional Form: Linearity

- We consider a model that is linear in the coefficients  $\alpha, \beta$ :  $Y_i = \alpha + \beta X_i + U_i$ .
- The dependent variable and the regressor can be nonlinear functions of some other variables.
- The most popular function is log.

## Log-Linear Specification

- Consider the following model:

$$\log Y_i = \alpha + \beta X_i + U_i.$$

- In this case,

$$\begin{aligned}\beta &= \frac{d(\log Y_i)}{dX_i} \\ &= \frac{dY_i/Y_i}{dX_i} = \frac{dY_i/dX_i}{Y_i}.\end{aligned}$$

- $\beta$  measures **percentage** change in  $Y$  as a response to a unit change in  $X$ .
- In this model, it is assumed that the percentage change in  $Y$  is the same for all values of  $X$  (constant).
- In  $\log(\text{Wage}_i) = \alpha + \beta \times \text{Education}_i + U_i$ ,  $\beta$  measures the **return** to education.

## Log-Log Specification

- Consider the following model:

$$\log Y_i = \alpha + \beta \log X_i + U_i.$$

- In this model,

$$\begin{aligned}\beta &= \frac{d \log Y_i}{d \log X_i} \\ &= \frac{dY_i/Y_i}{dX_i/X_i} = \frac{dY_i}{dX_i} \frac{X_i}{Y_i}.\end{aligned}$$

- $\beta$  measures **elasticity**: the percentage change in  $Y$  as a response to 1% change in  $X$ .
- Here, the elasticity is assumed to be the same for all values of  $X$ .
- Example: Cobb-Douglas production function:

$$Y = \alpha K^{\beta_1} L^{\beta_2} \implies \log Y = \log \alpha + \beta_1 \log K + \beta_2 \log L$$

(two regressors: log of capital and log of labour).

## Assumption of “Orthogonality”

The model:

$$Y_i = \alpha + \beta X_i + U_i.$$

We assume that  $E[U_i|X_i] = 0$ .

- $E[U_i] = 0$ .

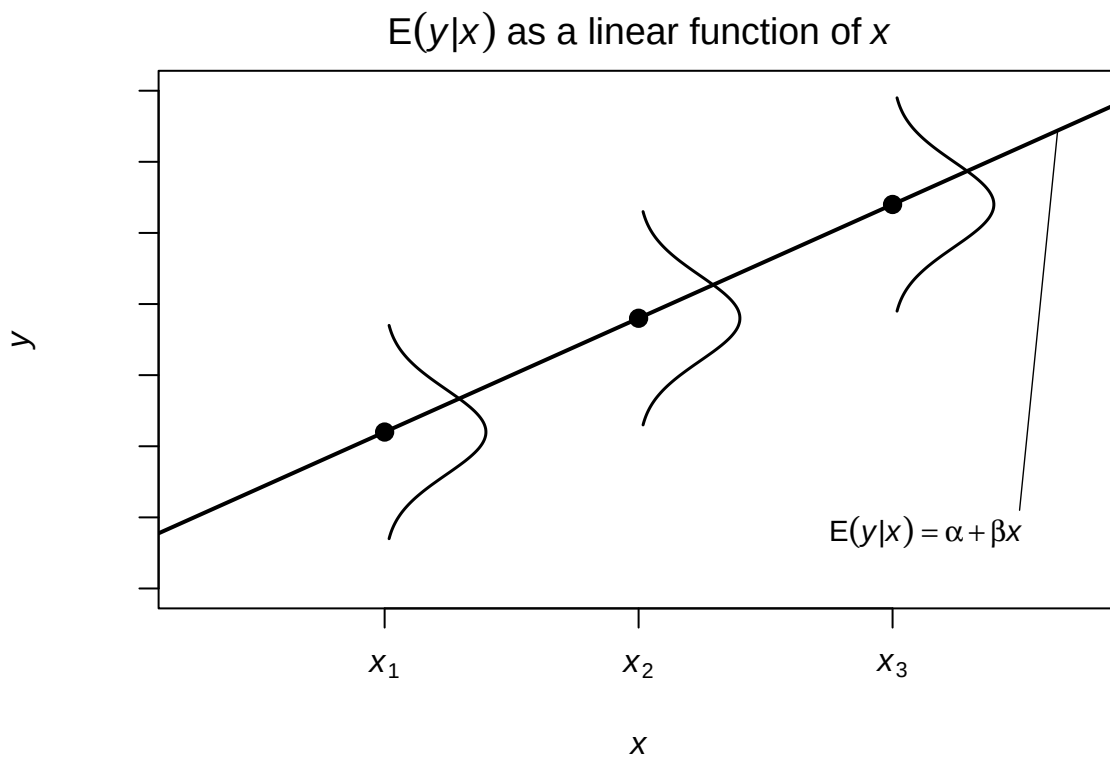
$$E[U_i] \stackrel{\text{Law of Iterated Expectation}}{=} E[E[U_i|X_i]] = E[0] = 0.$$

- $\text{Cov}(X_i, U_i) = E[X_i U_i] = 0$ .

$$\begin{aligned}E[X_i U_i] &\stackrel{\text{Law of Iterated Expectation}}{=} E[E[X_i U_i|X_i]] \\ &= E[X_i E[U_i|X_i]] = E[X_i \cdot 0] = 0.\end{aligned}$$

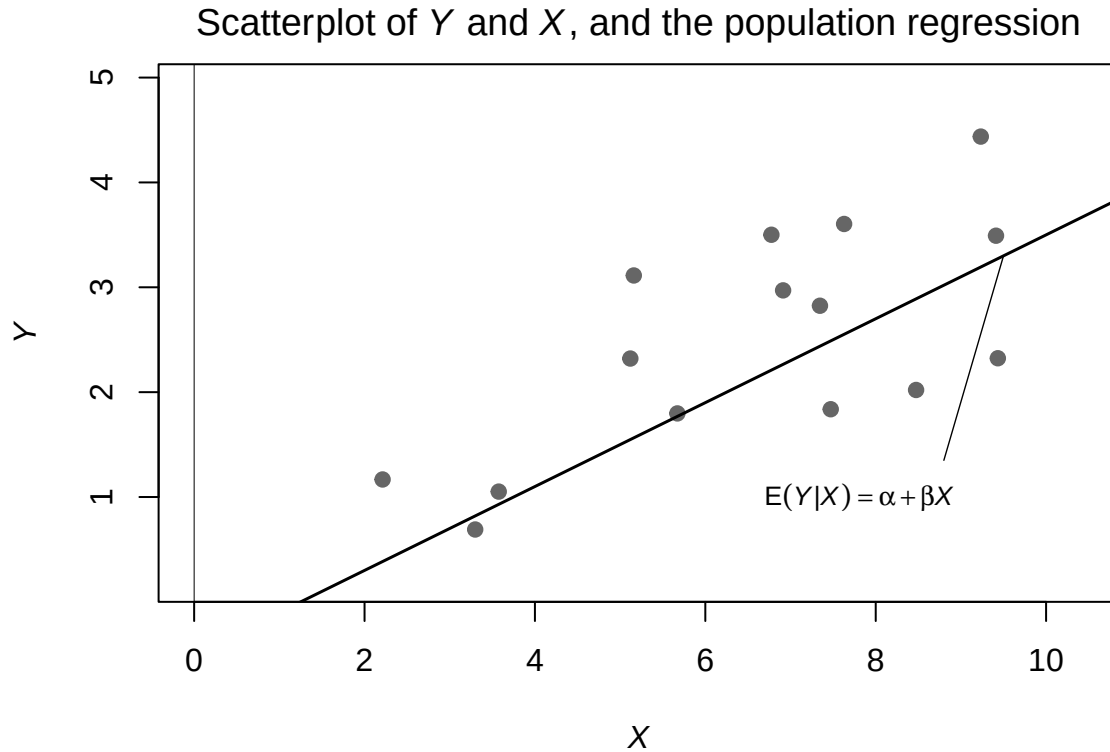
## Decomposition into Predicted and Error Components

$$Y_i = \underbrace{\alpha + \beta X_i}_{\text{Predicted by } X} + \underbrace{U_i}_{\text{"Orthogonal to" } X}$$



## Estimation of Parameters

### Estimation Problem



**Problem:** estimate the unknown parameters  $\alpha$  and  $\beta$  using the data ( $n$  observations) on  $Y$  and  $X$ .

## Method of Moments Estimator

- Our assumptions imply:

$$\begin{aligned} E[U_i] &= E[Y_i - \alpha - \beta X_i] = 0. \\ E[X_i U_i] &= E[X_i(Y_i - \alpha - \beta X_i)] = 0. \end{aligned}$$

- An **estimator** is a function of the observable data; it can depend only on observable  $X$  and  $Y$ . Let  $\hat{\alpha}$  and  $\hat{\beta}$  denote the estimators of  $\alpha$  and  $\beta$ .
- **Method of moments:** replace expectations with averages. **Normal equations:**

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i) &= 0. \\ \frac{1}{n} \sum_{i=1}^n X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) &= 0. \end{aligned}$$

## Normal Equations

- Let  $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$  and  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$  (averages).

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

implies

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n Y_i - \frac{1}{n} \sum_{i=1}^n \hat{\alpha} - \hat{\beta} \frac{1}{n} \sum_{i=1}^n X_i &= 0 \text{ or} \\ \bar{Y} - \hat{\alpha} - \hat{\beta} \bar{X} &= 0. \end{aligned}$$

The fitted regression line goes through the averages.

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$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}.$$

## Slope Estimator Derivation

- $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$  and therefore

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^n X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) \\ &= \sum_{i=1}^n X_i (Y_i - (\bar{Y} - \hat{\beta} \bar{X}) - \hat{\beta} X_i) \\ &= \sum_{i=1}^n X_i [(Y_i - \bar{Y}) - \hat{\beta} (X_i - \bar{X})] \\ &= \sum_{i=1}^n X_i (Y_i - \bar{Y}) - \hat{\beta} \sum_{i=1}^n X_i (X_i - \bar{X}). \end{aligned}$$

## Final Formulas for OLS Estimators

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$$0 = \sum_{i=1}^n X_i (Y_i - \bar{Y}) - \hat{\beta} \sum_{i=1}^n X_i (X_i - \bar{X}) \text{ or}$$

$$\hat{\beta} = \frac{\sum_{i=1}^n X_i (Y_i - \bar{Y})}{\sum_{i=1}^n X_i (X_i - \bar{X})}.$$

- Since

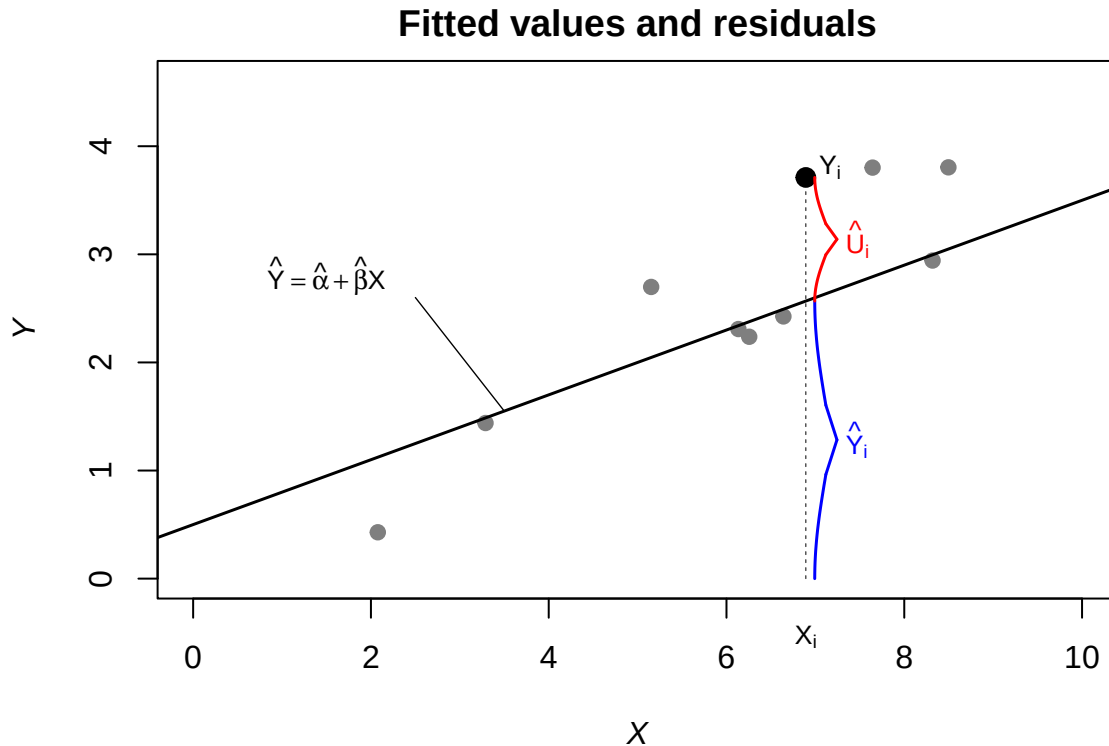
$$\begin{aligned} \sum_{i=1}^n X_i (Y_i - \bar{Y}) &= \sum_{i=1}^n (X_i - \bar{X}) (Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X}) Y_i \\ \sum_{i=1}^n X_i (X_i - \bar{X}) &= \sum_{i=1}^n (X_i - \bar{X}) (X_i - \bar{X}) = \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

we can also write

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

### Fitted Values and Residuals

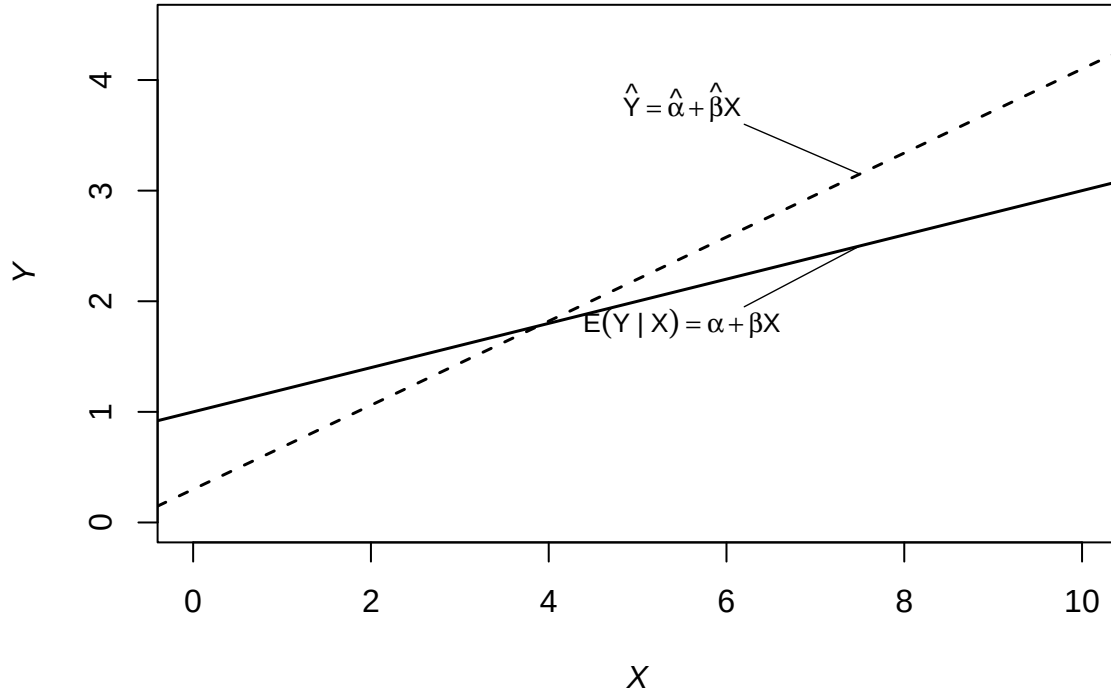
- **Fitted values:**  $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$ .
- **Fitted residuals:**  $\hat{U}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$ .



### True vs Fitted Regression Lines

- **True:**  $Y_i = \alpha + \beta X_i + U_i$ ,  $E[U_i] = E[X_i U_i] = 0$ .
- **Fitted:**  $Y_i = \hat{\alpha} + \hat{\beta}X_i + \hat{U}_i$ ,  $\sum_{i=1}^n \hat{U}_i = \sum_{i=1}^n X_i \hat{U}_i = 0$ .

## True vs fitted regression lines



## Ordinary Least Squares (OLS) Minimization

- Minimize  $Q(a, b) = \sum_{i=1}^n (Y_i - a - bX_i)^2$  with respect to  $a$  and  $b$ .
- Derivatives:

$$\frac{dQ(a, b)}{da} = -2 \sum_{i=1}^n (Y_i - a - bX_i).$$

$$\frac{dQ(a, b)}{db} = -2 \sum_{i=1}^n (Y_i - a - bX_i) X_i.$$

- First-order conditions:

$$0 = \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i) = \sum_{i=1}^n \hat{U}_i.$$

$$0 = \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i) X_i = \sum_{i=1}^n \hat{U}_i X_i.$$

- Method of moments = OLS:

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad \text{and} \quad \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}.$$